CCCA Meta-Atlas · Cell-State Manifold Analysis

Li et al. 2019 · Melanoma (Skin) · MARS-seq · 1,082 malignant cells (3 samples ≥ threshold)

GO / NO-GO GATES

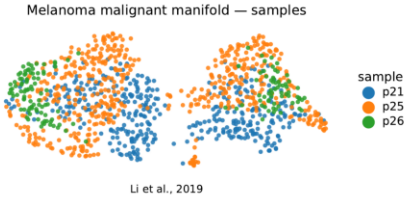
Gate 1 **GO** Mixed & continuous. At Harmony θ=40 the 3 samples share ONE connected component (0 detached cells), sample- and cell-cycle-mixed throughout. The 2-island split at θ≤20 is a technical MARS-seq low-complexity axis (minor island 11/12 markers are MT-pseudogenes / lincRNAs) — merged by stronger integration, not deleted.

Gate 2 **GO** Embryonic pattern present (non-branching). pSom/Gast co-localize at the origin with organized Gast↔pSom anti-correlation (r=−0.61); Crest is an inverse terminal gradient (Crest↔pSom r=−0.59) peaking at the terminus (decile pseudotime 0.80), NOT on a Gast arm — no inversion flag. No bifurcation.

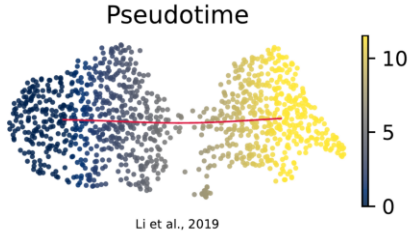
Gate 3 **NA** Not applicable — the manifold is non-branching (scFates curve: 2 tips, 0 forks, fold self-approach 2.98; a PPT tree adds no stable fork). With a single axis and no bifurcation there is no hub to test for mutual Gast↔pSom cross-repression, so Gate 3 is NA (branching-only).

STEP FIGURES

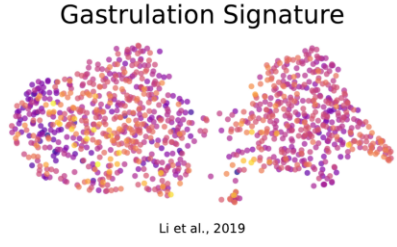
Samples (integrated manifold)



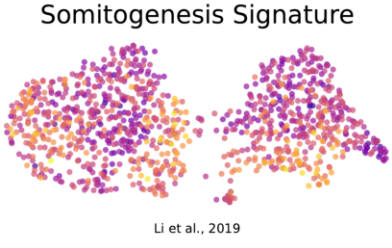
Pseudotime + curve



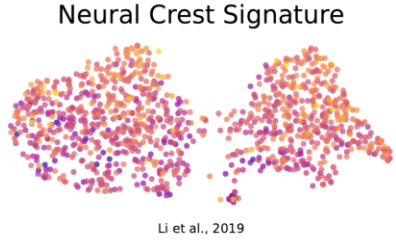
Gastrulation



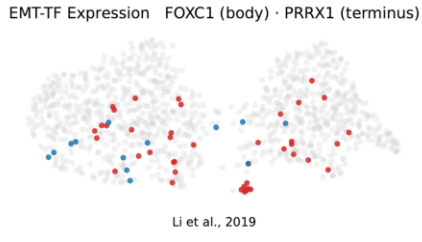
Somitogenesis



Neural crest



EMT-TF expression



PARAMETER CHOICES & RATIONALE

Min-cell threshold 1160 → 1082 cells. Gap-finder: sorted per-sample counts show a dominant spillover cliff 154→21 (ratio 7.3); geometric-mean threshold = 56 keeps p25/p21/p26 (154–528 cells), drops 17 spillover samples (≤8% purity floor).

Integration (θ) Harmony θ=40 on sample — first strength giving 1 continuous component with 0 detached cells (θ≤20 leaves 2 islands on a technical axis; θ=80 over-fragments). n_pcs=20 (precedent-bounded). Chosen on continuity, not a scalar.

Geometry / root Non-branching → scFates curve (not tree). Root = Gast/pSom-high, Crest-low tip (max Gast+pSom−Crest); Crest is developmentally late, never the root. Curve is fold-free (self-approach 2.98) and pseudotime-consistent (corr 0.99).

Module means (raw) Gast −0.015 · pSom +0.016 · Crest −0.002 — low-pSom, Crest-terminal melanoma (Linear-pSom/Crest geometry).

EMT-TFs Wide panel (Snail/Zeb/Twist + FOXC/PRRX; VIM excluded — marker, not a TF). Expressed: FOXC1 (body/origin) · PRRX1 (terminus). Categorical overlay of thresholded expressers on grey manifold.

VERDICT The embryonic program REPRODUCES as a non-branching axis — a single gastrulation/somitogenesis → neural-crest developmental gradient (G1 GO / G2 GO / G3 NA), the expected melanoma (neural-crest lineage) pattern.